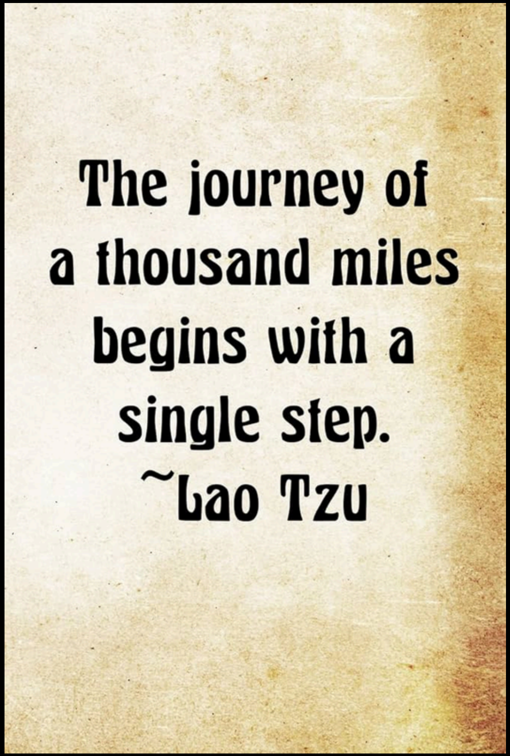
INTRODUCTION TO C++

The class will start at 9:05 PM



PLAN FOR TODAY

- How computers think?
- Why programming languages exist?
- Writing our first C++ code
- Rules of printing
- Data types & operators
- Taking input
- Making decisions (if-else)
- Assignments walkthrough

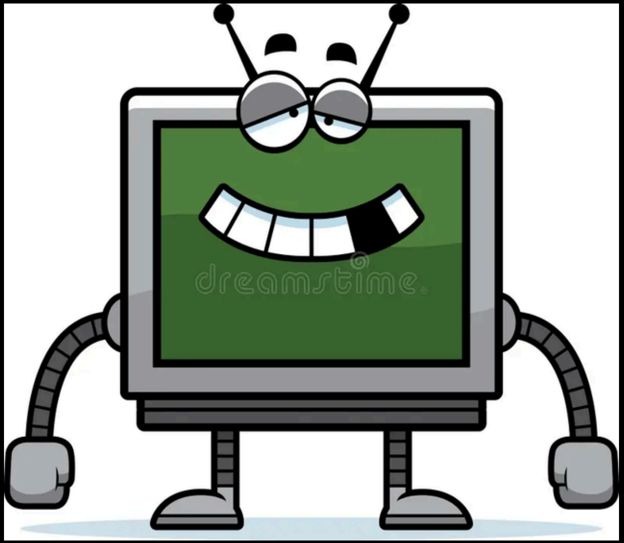
WHAT IS PROGRAMMING?

$\boxed{\text{Dumb}}$

↳ No common sense

↳ Can't think independently

↳ Follow if you provide proper instructions



Clear Instructions to computer, so that it can do some work for us.

WHY PROGRAMMING LANGUAGES EXIST?

Computers only understand 0's and 1's. (Binary)

Humans (English) ⟶ Program ⟶ Computers (Binary)

IDE & IMPORTANCE OF SYNTAX

FIRST C++ CODE (HELLO WORLD)

```
#include<bits/stdc++.h>
using namespace std;

int main()
{
    cout << "Hello world";
}
```

iostream

SOME IMPORTANT RULES TO REMEMBER

- Semicolon: Every statement must end with ;
- Case Sensitive: cout will work, but COUt will not.
- Printing Text: Text must be inside double quotes.
- New Line: For a new line, use endl.
- Brackets Come in Pairs

ARITHEMATIC OPERATORS

Operator	Meaning
+	Add
-	Subtract
*	Multiply
/	Divide
%	Remainder

Variables

A variable is a named box in the computer's memory where we store some data.

DATA TYPES (FOUNDATION)

Type	Use
`int`	Whole numbers
`long long`	Big numbers
`double`	Decimal
`char`	Single character
`bool`	true / false

SOME IMPORTANT RULES TO REMEMBER

- Variable must be declared before use
- No spaces in variable names
- Case sensitive: marks ≠ Marks
- Name should be meaningful (not x1, a2 everywhere)

TAKING INPUT

```
cin >> ____ >> ____ >> ____ ;
```

RELATIONAL OPERATORS

Operator	Meaning	Example	Result
`>`	Greater than	`5 > 3`	true
`<`	Less than	`5 < 3`	false
`>=`	Greater than or equal to	`5 >= 5`	true
`<=`	Less than or equal to	`4 <= 3`	false
`==`	Equal to	`5 == 5`	true
`!=`	Not equal to	`5 != 3`	true

int a = 10;

5 == 5 ✓

6 == 5 ✗

True / False

0 => False
Non Zero => True

```
bool ans = 5 == 5;

bool ans2 = 6 != 5;

bool ans3 = 6 < 5;

cout << ans << endl; // true
cout << ans2 << endl; // true
cout << ans3 << endl; // false
```

$\boxed{true}$
ans

$\boxed{true}$
ans2

$\boxed{false}$
ans3

LOGICAL OPERATORS

___ and ___	&&
___ or ___	||
not	!

CONDITIONALS

Blind

age $\geq$ 18

username

password

if "rainy"

carry umbrella

else

don't carry umbrella

if age $\geq$ 18

access

else

can't give access

$\boxed{\text{if else ladder}}$

```
int N;
cin >> N;

if (N % 2 == 0)
{
    ______
}
else
```

N

Even   Odd

2

4

6

8

10

Anything divisible by 2 => even

```
else
{
        ________
}
```

$\underbrace{\text{Max and Min of 2 Numbers}}$

A = 10 Max = 10

B = 7 Min = 7

```
if ( A > B )
{
    cout << "Max = " << A << endl;
}
else
{
    cout << "Max = " << B << endl;
```

```
}

if ( A < B )
{
        cout << " Min = " << A << endl;
}
else
{
        cout << " Min = " << B << endl;
}
```

Max and Min of 3 Numbers

A = 10

B = 12

C = 7

and   &&

12 12 12

12 12 7

12 7 12

Think before coding

Max

A => A > B and A > C

B => B > A and B > C

else

C

```
if (        )
{
        =
}
else if (     )
{
        =
}
else if (   )
{
}
```

Input: length, breadth

Solution

```
int length;
int breadth;

cin >> length >> breadth;

cout << "Area = " << length * breadth << endl;
cout << "Perimeter = " << 2*(length + breadth) << endl;
```

a (int)

$\boxed{10}$